

Review

Emerging landscape of KRAS inhibitors in cancer treatment

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SUMMARY

Alterations in *KRAS*, *NRAS*, and *HRAS* occur in roughly 20% of patients with cancer, making RAS one of the most intensively studied oncogenic targets. The discovery of mutant-selective *KRAS*^{G12C} inhibitors has provided a proof-of-concept for RAS-directed therapies, heralding a new era in the treatment of RAS-driven cancers. Yet, the efficacy of first-generation *KRAS*^{G12C} inhibitors is limited by the rapid emergence of resistance. Novel classes of (K)RAS inhibitors with distinct mechanisms of action and broader target coverage hold promise to overcome resistance and extend the benefits of RAS-targeted therapies to a wider patient population. In this review, we summarize clinical evidence for *KRAS*^{G12C} inhibitors across tumor types and delineate key mechanisms of resistance. We further discuss the rapidly evolving landscape of next-generation (K)RAS inhibitors, with particular emphasis on their target selectivity, mechanisms of action, preliminary clinical efficacy, and the therapeutic opportunities and challenges inherent to each class.

INTRODUCTION

RAS gene family

Mutations in the *RAS* gene family (*KRAS*, *NRAS*, and *HRAS*) occur in approximately 20% of human cancers, affecting an estimated 3.4 million new patients worldwide each year.¹ RAS proteins are ubiquitously expressed in almost all cell types, with *KRAS* representing the most abundant isoform.² Consistently, *in vivo* studies have shown that full knockout of the *KRAS* gene in genetically engineered mouse models results in embryonic lethality, while *NRAS* and *HRAS* are not essential for normal mouse development and can be compensated by *KRAS*.^{3,4} However, the replacement of *KRAS* with *HRAS* has been shown to rescue the embryonic effects of *KRAS* knockout, and knockout of *KRAS* in an adult mouse does not produce any obvious defects, suggesting that individual RAS isoforms are dispensable for normal cell viability.^{5,6}

Structure and function of RAS proteins

RAS proteins consist of either 189 amino acids (*HRAS*, *NRAS*, and *KRAS*-4A) or 188 amino acids (*KRAS*-4B), sharing a highly conserved catalytic G-domain among isoforms. (Figure 1A). Beyond this conserved domain, a C-terminal hypervariable region contributes to differences in membrane association and functional specificity among the isoforms.⁷ Functionally, RAS proteins belong to the GTPase family, acting as binary molecular switches that cycle between an inactive GDP-bound OFF state and an active GTP-bound ON state. This cycling is closely regulated by guanine nucleotide exchange factors (GEFs) and GTPase-activating proteins (GAPs). GEFs such as SOS1 catalyze nucleotide exchange from the GDP-bound OFF state to

the GTP-bound ON state, while GAPs including NF1 convert RAS to its inactive OFF state by markedly increasing RAS intrinsic GTP hydrolysis rate.⁸ Physiologically, GEF-mediated RAS activation is primarily triggered by external stimuli, most notably the binding of various growth factors to their extracellular receptors, which initiates an Receptor Tyrosine Kinase (RTK)-driven signaling cascade.⁹ The tyrosine phosphatase SHP2 facilitates RAS activation by dephosphorylating inhibitory sites and recruiting the GRB2-SOS1 complex to activated RTKs, while also mediating broader signaling through JAK-STAT, PI3K-AKT, and immune regulatory pathways.^{10–12} Once activated, RAS proteins undergo a conformational change of the switch I and switch II domains, enabling them to interact with multiple downstream effectors, most notably different RAF protein kinases such as RAF1, which in turn induces a cascade of phosphorylation reactions culminating in the activation of the canonical RAF/MEK/ERK pathway. While recent studies suggest that RAS driven cancers primarily depend on ERK signaling, other well established RAS effectors include PI3K, linking RAS with the PI3K/AKT/mTOR cascade, and RalGDS, which promotes RAS-like protein activation. Together, these signaling pathways mediate vital cellular functions including cell growth, proliferation, migration, metabolism, and differentiation.^{13,14}

Oncogenic mutations and their impact on RAS signaling

In the presence of oncogenic *RAS* mutations or gene amplifications, the closely regulated cycling of RAS is disrupted, leading to uncontrolled downstream signaling activation and cell growth. More than 90% of *RAS* mutations occur at codons 12, 13, and 61 (Figure 1B), causing a conformational change in the catalytic G domain of RAS proteins that limits GAP-mediated GTP



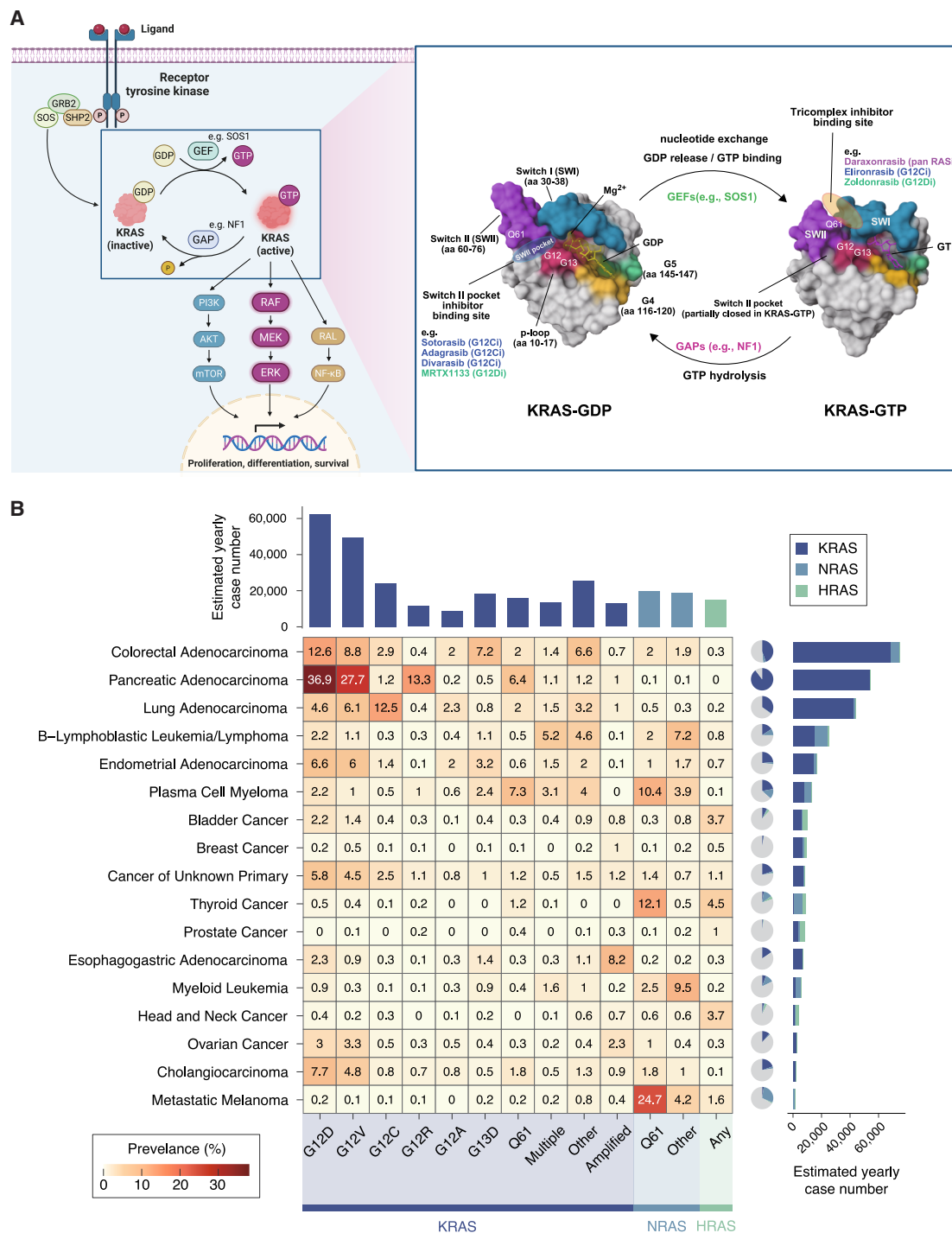


Figure 1. RAS structure, signaling, and oncogenic prevalence across tumor types

(A) RAS signaling pathway and structure of GDP and GTP bound KRAS, highlighting common mutation sites. KRAS structural motifs (P loop [residues 10–17], switch I [30–38], switch II [60–76], G4 [116–120], G5 [145–147]), and common oncogenic mutation sites (G12, G13, and Q61) are shown. KRAS cycles between the inactive GDP-bound OFF and active GTP-bound ON states, regulated by guanine nucleotide exchange factors (GEFs, e.g., SOS1) and GTPase-activating proteins (GAPs, e.g., NF1). GTP-bound KRAS activates downstream pathways, including RAF-MEK-ERK and PI3K-AKT, driving cell proliferation and survival. Switch II pocket inhibitors (e.g., sotorasib, adagrasib, divarasil, and MRTX1133) target GDP-bound KRAS, while tri-complex inhibitors (e.g., daraxonrasib, elironrasib, and zoldonrasib) engage the gap between switch I and II to target GTP-bound KRAS.

(B) Prevalence of distinct KRAS mutations and amplification across multiple cancer types. The heatmap shows the relative frequencies (%) of KRAS, NRAS, and HRAS mutations across representative cancer types, based on data from the AACR Project GENIE ($n = 108,509$ patient samples analyzed). The x axis denotes

(legend continued on next page)

hydrolysis. Consequently, the ratio of GTP-bound to GDP-bound RAS increases dramatically, shifting the equilibrium toward the active, ON state of RAS.^{1,8} Notably, depending on the specific mutation, some *RAS* mutants retain varying degrees of GAP-independent intrinsic GTP hydrolysis activity, allowing them to retain partial auto-inactivation and continue cycling between the ON and OFF states. Among common *KRAS* mutants, *KRAS*^{G12C} exhibits the highest intrinsic GTP hydrolysis activity, a key property that facilitated the development of the first mutant-specific *KRAS*^{G12C} OFF state inhibitors.¹⁵

Clinical importance and challenges of targeting *KRAS*

Accounting for 90% of all *RAS* mutations, *KRAS* is the most frequently altered isoform and the most common oncogenic driver in human cancers overall. The highest prevalence of *KRAS* mutations is observed in pancreatic ductal adenocarcinoma (PDAC), colorectal cancer (CRC), and non-small cell lung cancer (NSCLC) at rates of 90%, 45%, and 34%, respectively (Figure 1B). Taken together, these three malignancies account for more than one-third of cancer-related deaths in the Western world, underscoring the significant impact of *KRAS*-driven cancers on public health.^{16,17} Additionally, within specific tumor types, *KRAS* mutations are widely associated with poor prognosis, a scarcity of additional drug targets, and reduced responsiveness to available targeted therapies.^{18,19} As a result, *KRAS* and, to a lesser extent, *NRAS* and *HRAS* mutations have been considered as critical targets for therapeutic intervention since the early days of precision oncology.

Breakthroughs in *KRAS*^{G12C} inhibition

For decades, all attempts to target *KRAS* mutations in patients, either directly or indirectly through upstream regulators or downstream effectors, proved ineffective or too toxic, leading to the widespread conception that *KRAS* mutations were undruggable.⁷ In 2013, the groundbreaking discovery of a drug-binding pocket on the surface of mutant RAS, known as the switch II pocket, enabled the development of the first preclinical compounds that selectively and with high affinity bind to the chemically reactive cysteine residue of the *KRAS*^{G12C} mutant, while sparing wild-type RAS proteins, which lack reactive cysteine residues. By forming a covalent bond, these agents lock the mutant *KRAS*^{G12C} protein in its GDP-bound OFF state, thereby effectively preventing downstream signaling activation.²⁰ These seminal findings paved the way for the development of the first mutant selective *KRAS*^{G12C} OFF state inhibitors to enter the clinic - sotorasib and adagrasib.^{21,22} In the meantime, both drugs have demonstrated clinical efficacy and tolerability in *KRAS*^{G12C}-mutant cancers, leading to their approval by the U.S Food and Drug Administration (FDA) and European Medicines Agency (EMA) either as single agent therapy (NSCLC) or combined with anti-EGFR antibodies (CRC). However, while response rates are encouraging, the clinical benefit for patients on *KRAS*^{G12C} inhibition is relatively short-lived. Importantly, different classes of

novel (K)RAS inhibitors have been recently developed and are currently under clinical evaluation.²³ These agents have the potential not only to overcome resistance mechanisms associated with *KRAS*^{G12C} OFF state inhibitors but also advance therapeutic strategies by broadening the clinical applicability of (K)RAS-targeted therapies to patients with non-G12C (K)RAS mutations. In this review, we systematically summarize the clinical evidence for *KRAS*^{G12C} OFF state inhibitors, delineate key molecular mechanisms underlying therapeutic resistance, and provide a detailed overview of the rapidly evolving landscape of novel *KRAS* inhibitors, focusing on their target selectivity, mechanisms of action, advantages and limitations, and preliminary clinical efficacy data.

CLINICAL EXPERIENCES WITH MUTANT SELECTIVE *KRAS*^{G12C} OFF STATE INHIBITORS

Following proof-of-principle that the direct pharmacologic targeting of *KRAS* is feasible, numerous mutant-selective *KRAS*^{G12C} OFF state inhibitors are currently under clinical investigation. These small-molecule inhibitors share a common mechanism of action, as they exploit the switch II pocket to selectively and covalently bind the inactive, GDP-bound conformation of the *KRAS*^{G12C} protein. While cumulative clinical evidence provides encouraging proof-of-concept for the efficacy of this drug class, the clinical activity varies across different tumor types, with different mechanisms of resistance limiting the duration of response to these agents. Notably, next-generation *KRAS*^{G12C} OFF state inhibitors with enhanced potency and selectivity have recently advanced into clinical trials, demonstrating promising activity in early-phase studies (Table 1).

Non-small cell lung cancer

KRAS^{G12C} accounts for approximately 40% of all *KRAS* mutations and is the most common variant in lung adenocarcinoma (Figure 1B).⁵⁹ As a result, early clinical development efforts for mutant-selective *KRAS*^{G12C} inhibitors have primarily focused on this tumor type. Sotorasib was the first mutant-selective *KRAS*^{G12C} inhibitor to enter clinical investigation. In the landmark phase I Codebreak 100 trial evaluating sotorasib in advanced pretreated solid tumors harboring *KRAS*^{G12C} mutations, the NSCLC subgroup, which comprised 46% of the study population, achieved an overall response rate (ORR) of 32.2% and a median progression-free survival (PFS) of 6.3 months with no dose-limiting toxicities observed.²⁴ Extended findings from the subsequent phase II study in NSCLC confirmed these clinical benefits and demonstrated a median overall survival of 12.5 months, ultimately leading to conditional approval of sotorasib for this indication by the FDA (2021) and EMA (2022).²⁵ In the confirmatory randomized phase III Codebreak 200 trial, sotorasib showed a modest but statistically significant improvement in the primary endpoint of PFS by 5 weeks, a more favorable safety profile, and superior patient-reported outcomes compared to its comparator docetaxel.^{28,60} While the lack of a

RAS isoforms (*KRAS*, *NRAS*, and *HRAS*), and the y axis lists the selected cancer types. The bar graph above the heatmap shows the estimated annual number of U.S. patients with each alteration, calculated by combining 2025 U.S. Cancer Statistics incidence rates with GENIE-derived mutation frequencies. The stacked bar graph on the right depicts the estimated annual number of U.S. patients harboring *KRAS*, *NRAS*, or *HRAS* mutations in each cancer type. The pie charts summarize the proportion of RAS alterations by isoform within each cancer type. Details of the analytic methods are provided in the Supplement (Document S1).

Table 1. Clinical efficacy of KRAS^{G12C} OFF state inhibitors

Drug	Trial	Phase	Primary Endpoint	Tumor type	Disease setting	Number of patients	DCR % (95%CI)	ORR % (95%CI)	mPFS months (95%CI)	mOS months (95%CI)	Reference
Single-agent treatment											
Sotorasib	Codebreak 100 (NCT03600883)	1	Safety	NSCLC	Advanced pretreated	59	88.1 (77.07–95.09)	32.2 (20.62–45.64)	6.3 (3.9–7.8)	/	Hong et al. ²⁴
				CRC	Advanced pretreated	42	73.8 (57.96–86.14)	7.1 (1.50–19.48)	4.0 (2.8–5.5)	/	
				Other	Advanced pretreated	28	75.0 (55.13–89.31)	14.3 (4.03–32.67)	/	/	
Sotorasib	Codebreak 100 (NCT03600883)	2	ORR	NSCLC	Advanced pretreated	126	80.6 (72.6–87.2)	37.1 (28.6–46.2)	6.8 (5.1–8.2)	12.5 (10.0–ne)	Skoulidis et al. ²⁵
Sotorasib	Codebreak 100 (NCT03600883)	2	ORR	CRC	Advanced pretreated	62	82.3 (70.5–90.8)	9.7 (3.6–19.9)	4.0 (2.8–4.2)	10.6 (7.7–15.6)	Fakih et al. ²⁶
Sotorasib	Codebreak 100 (NCT03600883)	1/2	ORR	PDAC	Advanced pretreated	38	84 (69–94)	21 (10–37)	4.0 (2.8–5.6)	6.9 (5.0–9.1)	Strickler et al. ²⁷
Sotorasib vs. Docetaxel	Codebreak 200 (NCT04303780)	3	PFS	NSCLC	Advanced pretreated	171 Vs. 174	82.5 (75.9–87.8) vs. 60.3% (52.7–67.7)	28.1 (21.5–35.4) vs. 13.2 (8.6–19.2)	5.6 (4.3–7.8) vs. 4.5 (3.0–5.7)	10.6 (8.9–14.0) vs. 11.3 (9.0–14.9)	de Langen et al. ²⁸
Adagrasib	KRYSTAL-1 (NCT03785249)	1	Safety	NSCLC	Advanced pretreated	15	100 (/)	53.3 (26.6–78.7)	11.1 (2.6–ne)	/	Ou et al. ²⁹
				CRC	Advanced pretreated	2	100 (/)	50.0 (1.3–98.7)	/	/	
				Other	Advanced pretreated	2	100 (/)	0 (0.0–84.2)	/	/	
Adagrasib	KRYSTAL-1 (NCT03785249)	2	ORR	NSCLC	Advanced pretreated	116	79.5 (70.8–86.5)	42.9 (33.5–52.6)	6.5 (4.7–8.4)	12.6 (9.2–19.2)	Jänne et al. ³⁰
Adagrasib	KRYSTAL-1 (NCT03785249)	2	ORR	CRC	Advanced pretreated	44	83.7 (69.3–93.2)	23.3 (11.8–38.6)	5.6 (4.1–8.3)	19.8 (12.5–23.0)	Yaeger et al. ³¹
Adagrasib	KRYSTAL-1 (NCT03785249)	2	ORR	PDAC	Advanced pretreated	21	81.0 (/)	33.3 (22.9–48.9)	5.4 (3.9–8.2)	8.0 (5.2–11.8)	Bekaii-Saab et al. ³²
				Biliary	Advanced pretreated	12	91.7 (/)	41.7 (15.2–72.3)	8.6 (2.7–11.3)	15.1 (8.6–ne)	
				Other GI	Advanced pretreated	12	83.3 (/)	16.7 (2.1–48.4)	8.5 (2.5–17.7)	14.0 (6.2–ne)	
				Gynecologic	Advanced pretreated	7	85.7 (/)	57.1 (18.4–90.1)	8.1 (1.4–ne)	19.6 (15.1–ne)	
				Other tumors	Advanced pretreated	5	100.0 (/)	40.0 (5.3–85.3)	8.2 (7.0–ne)	7.6 (5.6–ne)	
Adagrasib vs. Docetaxel	KRYSTAL-12 (NCT04685135)	3	PFS	NSCLC	Advanced pretreated	301 vs. 152	78 (/) vs. 59 (/)	32 (26.7–37.5) vs. 9 (5.1–15.0)	5.5 (4.5–6.7) vs. 3.8 (2.7–4.7)	/	Barlesi et al. ³³

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Table 1. Continued

Drug	Trial	Phase	Primary Endpoint	Tumor type	Disease setting	Number of patients	DCR % (95%CI)	ORR % (95%CI)	mPFS months (95%CI)	mOS months (95%CI)	Reference
Divarasib	NCT04449874	1	Safety	NSCLC	Advanced pretreated	65	88.9 (/)	55.6 (42.5–68.1)	13.8 (9.8–25.4)	/	Sacher et al. and Li et al. ^{34,35}
				CRC		55	89.1 (/)	29.1 (17.6–42.9)	5.6 (4.1–8.2)	/	
				Other		20	95 (/)	36.0% (/)	/	/	
Glecirasib	NCT05009329	2	ORR	NSCLC	Advanced pretreated	117	86.3 (78.7–92.0)	47.9 (38.5–57.3)	8.2 (5.5–13.1)	/	Shi et al. ³⁶
Glecirasib*	NCT05194995	1/2	Safety/ORR	CRC	Advanced pretreated	44	86.4 (/)	22.7 (/)	5.6 (4.1–7.0)	16.0 (8.8–26.3)	Li et al. ³⁷
Glecirasib*	NCT05009329 NCT05002270	1/2	Safety/ORR	PDAC	Advanced pretreated	28	96.4 (/)	46.4(/)	5.5 (1.2–13.1)	/	Li et al. ³⁸
				Other tumors		19	84.2 (/)	52.6 (/)	7.0 (1.1–15.2)	/	
IBI351	NCT05005234	2	ORR	NSCLC	Advanced pretreated	116	90.5 (83.7–95.2)	49.1 (39.7–58.6)	9.7 (5.6–11.0)	/	Zhou et al. ³⁹
IBI351*	NCT05005234 NCT05497336	1	ORR	CRC	Advanced pretreated	45	85.0 (70.2–94.3)	47.5 (31.5–63.9)	/	/	Yuan et al. ⁴⁰
Garsorasib	NCT05383898	2	ORR	NSCLC	Advanced pretreated	123	89 (82–94)	50 (41–59)	7.6 (5.6–9.7)	/	Li et al. ⁴¹
Garsorasib	NCT04585035	1/2	Safety/ORR	CRC	Advanced pretreated	26	92.3 (74.9–99.1)	19.2 (6.6–39.4)	5.5 (2.9–11.6)	13.1 (9.5–ne)	Ruan et al. ⁴²
D3S-001	NCT05410145	1	Safety	NSCLC	Advanced pretreated	45	KRAS ^{G12C} i naive: 100 (83.9–/) KRAS ^{G12C} i pretreated: 80.0 (56.3–94.3)	KRAS ^{G12C} i naive: 66.7 (43.0–85.4) KRAS ^{G12C} i pretreated: 20.0 (5.7–43.7)	/	/	Cho et al. ⁴³
				CRC		13	88.9 (51.8–99.7)	88.9 (51.8–99.7)	/	/	
				PDAC		4	100 (39.8–/)	75 (19.4–99.4)	/	/	
Olomorasib*	(LOXO-RAS-20001) NCT04956640	1/2	Safety/ORR	NSCLC	Advanced pretreated	87	KRAS ^{G12C} i naive: 91 (/) KRAS ^{G12C} i pretreated: 82 (/)	KRAS ^{G12C} i naive: 36 (/) KRAS ^{G12C} i pretreated: 41 (/)	KRAS ^{G12C} i naive: 7.1 (5.5–8.9) KRAS ^{G12C} i pretreated: 8.1 (5.6–15.6)	/	Heist et al. and Hollebecque et al. ^{44,45}
				CRC		32	84 (/)	9 (/)	/	/	
				PDAC		24	92 (/)	33 (/)	/	/	
				Biliary		10	80 (/)	50 (/)	/	/	
				Other GI		7	100 (/)	50 (/)	/	/	
Opnurasib*	KontraST-01 (NCT04699188)	1/2	Safety/ORR	NSCLC	Advanced pretreated	34	/	41.7 (/)	/	/	Cassier et al. ⁴⁶

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Table 1. Continued

Drug	Trial	Phase	Primary Endpoint	Tumor type	Disease setting	Number of patients	DCR % (95%CI)	ORR % (95%CI)	mPFS months (95%CI)	mOS months (95%CI)	Reference
MK-1084*	KANDLELIT-001 (NCT05067283)	1	Safety	NSCLC	Advanced pretreated	21	76 (53–92)	38 (18–62)	8 (4-ne)	/	Sacher et al. ⁴⁷ ;
				CRC		53	83 (70–92)	43 (30–58)	/		Kuboki et al. ⁴⁸
Sosimerasib*	ChiCTR2200059986	2	ORR	NSCLC	Advanced pretreated	145	87.6 (81.1–92.5)	52.4 (44.0–60.8)	7.2 (5.6-ne)	/	Zhong et al. ⁴⁹
Combination treatment											
Sotorasib + Panitumumab	CodeBreak 101 (NCT04185883)	1	Safety	CRC	Advanced pretreated	48	92.5 (79.6–98.4)	30 (16.6–46.5)	5.7 (4.2–7.7)	15.2 (12.5-ne)	Kuboki et al. ⁵⁰
Sotorasib 960mg + Panitumumab vs. Sotorasib 240mg + Panitumumab vs. SOC (Trifluridine-Tipiracil/Regorafenib)	CodeBreak 300 (NCT05198934)	3	PFS	CRC	Advanced pretreated	53 vs. 54	71.7 (57.7–83.2) vs. 69.8 (55.7–81.7) vs. 46.3 (32.6–60.4)	30.2 (18.3–44.3) vs. 7.5 (2.1–18.2) vs. 1.9 (0.0–9.9)	5.7 (4.2–7.5) vs. 4.01 (3.7–5.9) vs. 2.04 (1.9–3.9)	ne (8.61-ne) vs. 11.9 (7.5-ne) vs. 10.3 (7.0-ne)	Pietrantonio et al. ⁵¹
Adagrasib + Cetuximab	KRYSTAL-1 (NCT04793958)	1/2	Safety/ORR	CRC	Advanced pretreated	94	85.1 (76.3–91.6)	34.0 (24.6–44.5)	6.9 (5.7–7.4)	15.9 (11.8–18.8)	Yaeger et al. ⁵²
Divarasib + Cetuximab	NCT04449874	1	Safety	CRC	Advanced pretreated	29	KRAS ^{G12C} i naive: 100 (/) KRAS ^{G12C} i pretreated: 100 (/)	KRAS ^{G12C} i naive: 62.5 (40.6–81.2) KRAS ^{G12C} i pretreated: 60 (/)	KRAS ^{G12C} i naive: 8.1 (5.5–12.3) KRAS ^{G12C} i pretreated: /	/	Desai et al. ⁵³
Garsorasib + Cetuximab	NCT04585035	1/2	Safety/ORR	CRC	Advanced pretreated	42	92.9 (80.5–98.5)	45.2 (29.8–61.3)	7.5 (5.5–8.1)	nr (11.3-ne)	Ruan et al. ⁴²
Glecirasib* + Cetuximab	NCT05194995	1/2	Safety/ORR	CRC	Advanced pretreated	46	87 (/)	50 (/)	6.8 (5.4–6.9)	19.3 (13.1-ne)	Li et al. ³⁷
Olmorasin* + Cetuximab	LOXO-RAS-20001 (NCT04956640)	1/2	Safety/ORR	CRC	Advanced pretreated	93	92.5 (85.1–96.9)	43.0 (32.8–53.7)	7.4 (6.6–7.8)	/	Hollebecq et al. ⁵⁴
MK-1084* + Cetuximab	KANDLELIT-001 (NCT05067283)	1	Safety	CRC	Advanced pretreated	39	95 (83–99)	56 (40–72)	/	/	Lugowska et al. ⁴⁸

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Table 1. Continued

Drug	Trial	Phase	Primary Endpoint	Tumor type	Disease setting	Number of patients	DCR % (95%CI)	ORR % (95%CI)	mPFS months (95%CI)	mOS months (95%CI)	Reference
Sotorasib* + Panitumumab + FOLFIRI	CodeBreak 101 (NCT04185883)	1	Safety	CRC	Advanced pretreated	40	92.5 (76.1–98.4)	57.5 (40.9–73.0)	8.2 (7.0–10–8)	15.6 (12.9–23.3)	Strickler et al. ⁵⁵
MK-1084* + Cetuximab + FOLFOX	KANDLELIT-001 (NCT05067283)	1	Safety	CRC	Advanced untreated/ pretreated	29	92 (62–100)	38 (21–58)	/	/	Lugowska et al. ⁴⁸
Sotorasib* + Pembrolizumab/ Atezolizumab	CodeBreak 100/101 (NCT04185883)	1	Safety	NSCLC	Advanced pretreated	58	83 (71–91)	29 (18–43)	/	15.7 (9.8–17.8)	Li et al. ⁵⁶
Adagrasib* + Pembrolizumab	KRYSTAL-7 (NCT04613596)	2	ORR	NSCLC	Advanced untreated	149	/	PD-L1 expression >50%: 61.0 (46–74) PD-L1 expression 1–49%: 41.0 (25–58) PD-L1 expression <1%: 36.0 (23–50)	PD-L1 expression >50%: 27.7 (9.6-ne) PD-L1 expression 1–49%: 13.5 (2.3-ne) PD-L1 expression <1%: 8.2 (2.7–13.6)	PD-L1 expression >50%: NR (19.4-ne) PD-L1 expression 1–49%: 14.3 (8.3-ne) PD-L1 expression <1%: 15.5 (9.6-ne)	Jänne et al. ⁵⁷
Olomorasib* + Pembrolizumab	LOXO-RAS-20001 (NCT04956640)	1/2	Safety/ ORR	NSCLC	Advanced untreated	48	All comers: 91 (79.2–97.6) PD-L1 expression >50%: 905 (75.1–99.9)	All comers: 74 (58.9–85.7) PD-L1 expression >50%: 90 (68.3–98.8)	/	/	Dragnev et al. ⁵⁸

Conference presentations are indicated with an asterisk (*). Abbreviations: DCR – disease control rate; CI – confidence interval; ORR – objective response rate; mPFS – median progression free survival; mOS – median overall survival; Ref. – reference; NSCLC – non-small cell lung cancer; CRC – colorectal cancer; PDAC – pancreatic ductal adenocarcinoma; GI – gastrointestinal; ne – not estimable; nr – not reached; KRAS^{G12C}i – KRAS^{G12C} inhibitor.

significant overall survival (OS) benefit in the trial may be partly attributed to a high crossover rate to other KRAS^{G12C} inhibitors in the control arm, additional concerns have been raised regarding the trial design, including the selection of a suboptimal comparator (docetaxel monotherapy instead of the more effective docetaxel plus ramucirumab regimen) as well as the potential for investigator bias, such as the premature determination of progressive disease in the docetaxel arm.⁶¹ Thus, the FDA has demanded an additional confirmatory phase III trial to be completed by February 2028 to support full approval of sotorasib in this setting. In parallel, a randomized phase III trial comparing sotorasib with pembrolizumab in combination with platinum-based doublet chemotherapy as first-line treatment for advanced, PD-L1-negative NSCLC is ongoing, with primary completion expected in 2026 (NCT05920356).⁶²

Adagrasib, the second mutant-selective covalent KRAS^{G12C} OFF state inhibitor to enter clinical development, is functionally similar to sotorasib but features a substantially longer half-life.²² In the phase I/II KRYSTAL-1 trial, adagrasib demonstrated a manageable safety profile and robust anti-tumor activity in pretreated NSCLC showing a 43% ORR, a mPFS of 6.5 months, and a mOS of 12.6 months alongside an intracranial ORR of 33% in the subgroup of patients with stable central nervous metastases.³⁰ These results supported conditional approval by the FDA and EMA. In the recently published confirmatory phase III KRYSTAL-12 trial, adagrasib demonstrated a significant PFS advantage over docetaxel (median 5.5 vs. 3.8 months) and a delay in lung cancer-related symptom deterioration, with OS data still pending. Notably, while overall rates of treatment-related adverse events were similar between arms, adagrasib was associated with higher rates of gastrointestinal toxicities, leading to dose interruptions in 59% of patients.³³ Toxicities associated with adagrasib and other KRAS^{G12C} inhibitors, notably gastrointestinal effects, are mechanistically unclear and warrant further study.

Multiple other next-generation covalent KRAS^{G12C} OFF state inhibitors are currently under early phase clinical exploration in NSCLC and other malignancies, with the most mature data in Western patients existing for divarasib, a highly selective KRAS^{G12C} OFF state inhibitor with a 5- to 20-fold higher *in vitro* potency than sotorasib and adagrasib. In a recently published long-term follow-up of the NSCLC subgroup from a phase I basket study, divarasib demonstrated an ORR of 55.6% and a median PFS of 13.8 months, with no dose-limiting toxicities observed at the tested doses.³⁴ The randomized controlled phase III KRASCENDO-1 trial evaluating the efficacy of divarasib versus sotorasib or adagrasib in patients with pretreated advanced NSCLC is currently recruiting with primary completion expected in 2026 (NCT06497556).⁶³ Published and preliminary clinical data on other next-generation covalent KRAS^{G12C} OFF state inhibitors including glecirasib,³⁶ IBI351,³⁹ olomorasib,⁴⁴ garsorasib,⁴¹ D3S-001,⁴³ opnurasis,⁴⁶ MK-1084⁴⁷ and sosimerasib⁴⁹ are summarized in Table 1.

Based on compelling preclinical evidence that KRAS inhibitors synergize with immune checkpoint inhibitors, particularly in immunologically hot mouse models, by remodeling the immunosuppressive tumor microenvironment associated with KRAS mutations, multiple clinical trials are currently evaluating KRAS^{G12C} inhibitors in combination with immune checkpoint in-

hibitors in NSCLC.⁶⁴ The CodeBreak 101 trial, which evaluated sotorasib combined with either pembrolizumab or atezolizumab, was prematurely terminated due to severe treatment-related hepatotoxicity.⁵⁶ More encouraging data have recently been reported from the phase II KRYSTAL-7 trial on adagrasib plus pembrolizumab in treatment-naïve patients with advanced NSCLC. In a preliminary analysis of patients with a PD-L1 tumor proportion score greater than 50%, a subgroup generally considered immunotherapy-sensitive, adagrasib plus pembrolizumab demonstrated an ORR of 61% and a remarkable mPFS of 27.7 months, comparing favorably to historical outcomes with single-agent pembrolizumab in cross-trial comparisons. Although grade 3/4 treatment-related adverse events occurred in 65% of patients, severe hepatotoxicity was less frequent than observed with sotorasib in the CodeBreak 101 study, and the overall safety profile was considered manageable.^{57,65} Randomized phase III trial evaluating adagrasib (NCT04613596, NCT06875310)^{66,67} as well as divarasib (NCT06793215),⁶⁸ olomorasib (NCT06890598, NCT06119581)^{69,70} and MK-1084 (NCT07190248, m1e) in combination with pembrolizumab with or without chemotherapy as first-line therapy for advanced NSCLC, are currently enrolling. Primary completion for most of these studies is anticipated in 2028–2029.

A multitude of other combinatorial strategies aimed at enhancing KRAS^{G12C} OFF state inhibitor efficacy and durability are currently in early-phase clinical trials across KRAS^{G12C}-mutant NSCLC and other solid tumors, although mature clinical data remain scarce. Among the most promising approaches - discussed in greater detail elsewhere - are vertical inhibition strategies that co-target core components of the RAS/MAPK pathway by combining KRAS^{G12C} inhibitors with upstream modulators such as RTK, SOS1, or SHP2 inhibitors, or with agents targeting downstream effectors including RAF, MEK, and ERK. Additional concepts under investigation include co-targeting convergent pathways such as the PI3K/AKT/mTOR axis and reinforcing cell-cycle arrest via combinations with CDK4/6 inhibitors.⁷¹

Colorectal cancer

In contrast to NSCLC, single-agent KRAS^{G12C} OFF state inhibitors have demonstrated limited efficacy in metastatic CRC. In the CRC cohort of the phase II CodeBreak 100 trial, sotorasib yielded an ORR of 9.7% and a median PFS of 4.0 months.²⁴ While cross-trial comparisons suggest slightly higher ORRs for adagrasib (23%, KRYSTAL-1),³¹ divarasib (29.1%, NCT04449874)³⁵ and other emerging KRAS^{G12C} off-state inhibitors (Figure 2A; Table 1), clinical benefits remain transient, with median PFS estimates ranging from 5.6 to 7.8 months.^{31,35,37,44,72,73} Preclinical studies indicate that this limited efficacy in CRC may be driven, at least in part, by rapid RAS/MAPK pathway reactivation following KRAS^{G12C} inhibition, mediated by the adaptive upregulation of receptor tyrosine kinases (RTKs) such as the epidermal growth factor receptor (EGFR) that activate bypass signaling through wild-type RAS isoforms.⁷⁴ The observed synergy between KRAS^{G12C} inhibitors and EGFR inhibition in preclinical models,^{75,76} together with prior clinical success of vertical RAS/MAPK pathway blockade in BRAF^{V600E}-mutant CRC,⁷⁷ provided the rationale for the clinical investigation of combined KRAS^{G12C} inhibition and anti-EGFR antibody therapy in metastatic CRC. Initial clinical

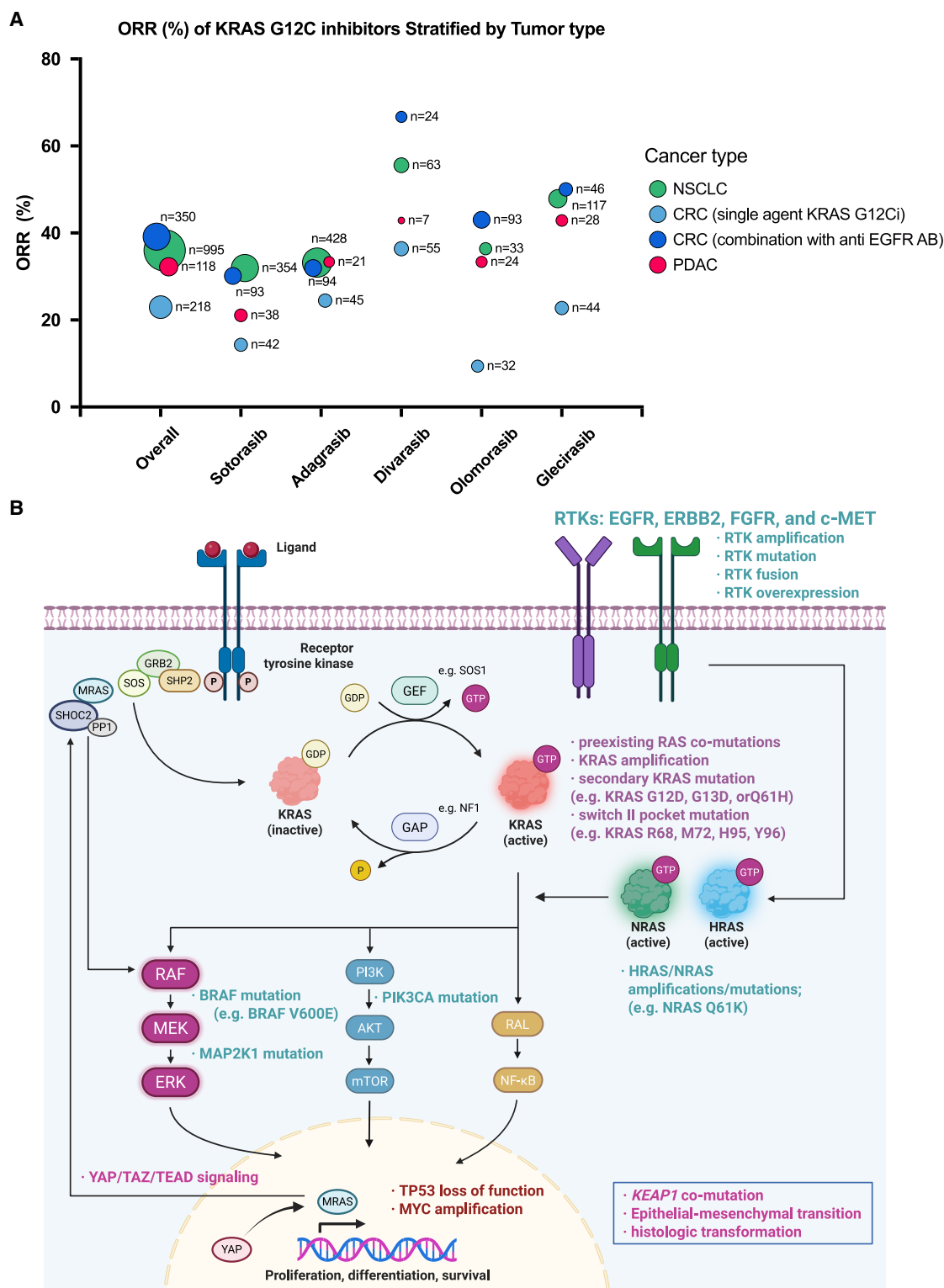


Figure 2. Clinical efficacy and resistance mechanisms of KRAS G12C OFF state inhibitors across cancer types

(A) Clinical activity of different KRAS^{G12C} OFF state inhibitors across tumor types. Bubble plot shows overall response rates (ORRs) of individual KRAS^{G12C} OFF state inhibitors (sotorasib, adagrasib, divarasiab, olomorasib, and glecirasib) stratified by tumor type, with “Overall” representing the combined data across all inhibitors. Data are presented only for inhibitors with ORR data available across all four indications: NSCLC, CRC (single-agent KRAS^{G12C} inhibitor), CRC (in combination with anti-EGFR antibody), and PDAC. Patients previously treated with a KRAS^{G12C} inhibitor were excluded from the analysis. Bubble size corresponds to the number of patients.

trial data on this combination emerged from the phase 1b cohort of the KRYSTAL-1 study, evaluating adagrasib in combination with the anti-EGFR monoclonal antibody cetuximab in patients with pretreated metastatic CRC. In a non-randomized comparison with single-agent adagrasib, updated data on adagrasib plus cetuximab demonstrated a markedly improved ORR of 34%, a median PFS of 6.9 months, and a median OS of 15.9 months, with a manageable safety profile.^{31,52} These results supported the accelerated FDA approval of adagrasib plus cetuximab for chemotherapy refractory metastatic CRC. In parallel, the randomized phase III KRYSTAL-10 trial is evaluating adagrasib plus cetuximab versus standard of care as second-line therapy in advanced CRC and has completed enrollment (NCT04793958).⁷⁸ Consistent with its potent single agent activity, divarasil in combination with cetuximab demonstrated the highest ORR reported to date for a KRAS^{G12C} OFF-state inhibitor plus anti-EGFR antibody combination in metastatic CRC. In the phase Ib NCT04449874 trial, including patients with pretreated metastatic CRC, divarasil combined with cetuximab was well tolerated and achieved a promising ORR of 62.5%, including partial responses in 3 of 5 patients who had previously received a KRAS^{G12C} inhibitor.⁵³

The first randomized trial data on KRAS^{G12C} inhibitor plus anti-EGFR antibody treatment were presented in the phase 3 Codebreak 300 trial. Sotorasib at a dose of 960 mg daily in combination with panitumumab demonstrated a statistically significant improvement of the primary endpoint with a median PFS of 5.6 months versus 2.0 months with standard-of-care treatment (trifluridine-tipiracil or regorafenib) alongside an ORR of 30.2%, leading to its FDA approval in patients with chemotherapy refractory metastatic CRC. Although the updated analysis of the Codebreak 300 trial did not demonstrate a statistically significant difference in OS—an endpoint for which the study was underpowered—a clear trend toward improved OS was observed with sotorasib plus panitumumab supporting its clinical use in this setting.⁵¹ Preliminary results from the phase Ib Codebreak 101 trial of sotorasib plus panitumumab in combination with FOLFIRI as first-line therapy in KRAS^{G12C}-mutant mCRC demonstrated a manageable safety profile and an ORR of 78%, with tumor shrinkage observed in all evaluable patients.⁷⁹ Randomized phase III trials evaluating first-line sotorasib plus panitumumab and chemotherapy (NCT06252649),⁸⁰ as well as MK-1084 combined with cetuximab and chemotherapy (NCT06997497), in patients with advanced CRC are ongoing. Collectively, these developments suggest that mutant selective off-state KRAS^{G12C} inhibitors may soon be integrated into early-line and potentially (neo)adjuvant treatment strategies for CRC.

Pancreatic ductal adenocarcinoma and other malignancies

Although KRAS^{G12C} OFF state inhibitors have so far been approved only for NSCLC and CRC—tumor types with the high-

est prevalence of KRAS^{G12C} mutations—emerging data suggest potential tumor-agnostic activity beyond these indications. In the Codebreak 100 trial, a subgroup analysis reported an ORR of 14.3% among patients with KRAS^{G12C}-mutant tumors other than NSCLC and CRC.²⁴ Notably, in heavily pretreated patients with advanced PDAC—a population with dismal prognosis and very limited treatment options—sotorasib achieved an ORR of 21% and a median PFS of 4.0 months.²⁷ Slightly better efficacy was observed in a cross-trial comparison with adagrasib, demonstrating an ORR of 35.1% in patients with various KRAS^{G12C}-mutant tumors, including responses in PDAC (ORR: 33.3%, mPFS: 5.4 months), biliary tract cancer (ORR: 41.7%, mPFS: 8.6 months), and gynecologic malignancies (ORR: 57.1%, mPFS: 8.1 month).³² Encouraging preliminary efficacy data in PDAC and other malignancies have also been reported for several next-generation KRAS^{G12C} OFF state inhibitors, including divarasil, glecirasil, opnurasil, olomorasil, and IBI351, as summarized in Table 1.

Efficacy of KRAS^{G12C} OFF state inhibitors across different tumor types

Interestingly, among the three major cancer types harboring KRAS mutations—NSCLC, CRC, and PDAC—KRAS^{G12C} OFF state inhibitors have demonstrated a consistent pattern of differential sensitivity between tumor types, suggesting key lineage-specific differences in adaptive resistance mechanisms and/or KRAS dependency (Figure 2A). In detail, ORR tends to be highest in NSCLC, with PDAC only slightly lower, but with response rates in CRC nearly half that of NSCLC. Fortunately, consistent with the known role of EGFR-driven adaptive feedback reactivation of RAS-MAPK signaling in CRC, combinations of KRAS^{G12C} inhibitors with anti-EGFR antibodies have, in all cases, produced ORR comparable to those seen in NSCLC, suggesting that the optimal targeting of KRAS can lead to favorable activity across all three tumor types. These observations also suggest that differences in sensitivities between tumor types might also be observed with newer classes of KRAS inhibitors.

MECHANISMS OF RESISTANCE TO KRAS^{G12C} OFF STATE INHIBITORS

While the successful clinical implementation of KRAS^{G12C} OFF state inhibitors has marked the start of a new era of RAS-targeted cancer therapy, clinical benefit remains modest, with most patients experiencing disease progression within months due to the rapid emergence of resistance. Similar to previous breakthroughs in targeted cancer therapy, such as BCR-ABL inhibition in chronic myeloid leukemia and EGFR targeting in NSCLC, delineating and functionally characterizing these resistance mechanisms is critical to inform the development of next-generation inhibitors and rational combination strategies capable of delaying or preventing therapeutic escape.^{81,82}

(B) Mechanisms of resistance to KRAS G12C OFF state inhibitor. Schematic overview of signaling pathways and genetic alterations that mediate intrinsic or acquired resistance to KRAS^{G12C} OFF state inhibitors. Upstream receptor tyrosine kinase (RTK) activation through amplification, mutation, fusion, or overexpression (e.g., EGFR, ERBB2, FGFR, and c-MET) can restore RAS pathway signaling. On-target mechanisms include KRAS amplification, secondary mutations (e.g., G12D, G13D, and Q61H), or switch II pocket mutations (e.g., R68, M72, H95, and Y96). Resistance can also arise from alterations in other RAS isoforms, including HRAS/NRAS amplification or activating mutations (e.g., NRAS Q61K). Downstream alterations encompass activating mutations in BRAF (e.g., V600E), MAP2K1, or PIK3CA, as well as TP53 loss, MYC amplification, or KEAP1 co-mutations. Non-genetic mechanisms include epithelial-mesenchymal transition, histologic transformation, and activation of bypass pathways such as YAP/TAZ/TEAD signaling.

Mechanisms of resistance to KRAS^{G12C} OFF state inhibitors have been extensively reviewed elsewhere.^{83,84} Here, we focus on the key principles necessary to better understand the promises and potential pitfalls of next-generation (K)RAS inhibitors. From a clinical perspective, resistance to KRAS^{G12C} inhibition can be broadly divided into two categories. Primary resistance refers to preexisting tumor-intrinsic features that render a tumor inherently unresponsive to KRAS^{G12C} inhibition. In contrast, acquired resistance arises during the course of treatment in initially sensitive tumors and ultimately leads to disease progression. Functionally, KRAS^{G12C} inhibitor resistance can be further stratified into genetic and non-genetic mechanisms, which are thought to coevolve under therapeutic pressure (Figure 2B).

Primary resistance

While KRAS^{G12C} off-state inhibitors induce transient tumor control in most patients, primary resistance occurs in 10–30% of cases (see DCR in Table 1). Beyond the presence of the KRAS^{G12C} mutation itself, validated predictive biomarkers to preselect for likely responders remain scarce. A subgroup analysis from the NCT04449874 trial evaluating divarasib in patients with KRAS^{G12C}-mutant solid tumors has shown that patients harboring additional preexisting RAS mutations did not respond to therapy.³⁵ Consistently, preclinical studies have shown that secondary RAS mutations confer resistance to various first generation KRAS^{G12C} OFF state inhibitors by reactivating the RAS/MAPK pathway. However, because such co-existing KRAS alterations are typically very rare in targeted therapy naive tumors, their utility as predictive biomarkers for upfront patient selection remains limited.⁸⁵ In NSCLC, a large-scale biomarker analysis of baseline tumor samples from over 400 patients treated with the KRAS^{G12C} inhibitors sotorasib or adagrasib revealed that co-occurring mutations in the tumor suppressor genes *KEAP1*, *SMARCA4*, and *CDKN2A*—present in more than one-quarter of patients—are associated with inferior clinical outcomes. Among these, loss-of-function mutations in *KEAP1* have consistently emerged across multiple datasets as a strong predictor of poor response to KRAS^{G12C} inhibition.^{86–88} These clinical observations are further supported by preclinical studies indicating that *KEAP1* inactivation promotes resistance to KRAS-targeted therapy, likely through enhanced NRF2-mediated transcriptional programs that sustain tumor cell survival.⁸⁷ Notably, *KEAP1* mutations have also been associated with inferior outcomes following immune checkpoint blockade in NSCLC, raising the question of whether these alterations primarily reflect a broader role in promoting aggressive tumor biology and poor prognosis, independent of therapeutic context.⁸⁹ This hypothesis is further supported by a recently published comprehensive biomarker analysis from the randomized phase 3 Codebreak 200 trial, demonstrating that *KEAP1* mutational status was not predictive of differential PFS benefit from sotorasib versus docetaxel. Interestingly, *ATM* mutational status emerged as the only predictive biomarker in this analysis, with *ATM* wild-type patients deriving a significantly greater PFS benefit from sotorasib.⁸⁸

Beyond NSCLC, data on mechanisms of primary resistance to KRAS^{G12C} inhibitors remain limited. In KRAS^{G12C}-mutant CRC and PDAC, *TP53* and *PIK3CA* are among the most frequently co-mutated genes. Both have been implicated in potentially driving resistance to KRAS^{G12C} inhibition, particularly in preclin-

ical models. However, a subgroup analysis from the non-randomized KRYSTAL-1 trial evaluating adagrasib in combination with cetuximab in KRAS^{G12C}-mutant mCRC did not reveal a correlation between baseline *TP53* or *PIK3CA* co-mutations and treatment response. These findings suggest that the clinical relevance of these alterations in predicting primary resistance remains uncertain and highlight the need for further mechanistic and clinical investigation.

Genetic mechanisms of acquired resistance

The genomic landscape of emergent alterations conferring resistance to KRAS^{G12C} inhibitors has been increasingly well characterized.^{35,53,85,90–93} This progress has been largely driven by the widespread implementation of highly sensitive next-generation sequencing of circulating tumor DNA (ctDNA) in clinical trials, enabling minimally invasive, real-time detection of candidate drivers of resistance during treatment and at the time of clinical progression. Yet, it is important to recognize that ctDNA analysis is inherently limited to detecting genetic resistance mechanisms only and is constrained by factors such as low tumor DNA shedding in certain cancers and difficulties in detecting low frequency subclonal mutations. Moreover, the identification of newly emergent alterations in ctDNA does not necessarily indicate functional significance and requires validation through preclinical models to distinguish true resistance drivers from passenger alterations.⁹⁴

Across different tumor types, acquired genetic alterations are detected in roughly 60% of patients at the time of clinical resistance to KRAS^{G12C} inhibitors, with the majority harboring at least one newly emergent alteration in the canonical RAS/MAPK pathway.^{84,90} These findings underscore the continued oncogenic dependency of KRAS-mutant cancers on sustained RAS/MAPK signaling and supports the rationale for next-generation (K)RAS inhibitors with broader target engagement, as well as combination strategies employing vertical pathway inhibition. The genetic landscape of acquired resistance in patients treated with KRAS^{G12C} OFF state inhibitors can be broadly categorized into two classes: on-target alterations that directly affect the KRAS protein itself, and off-target alterations that reactivate downstream or bypass signaling. On-target KRAS alterations are detected in approximately 40% of patients at the time of clinical progression and can be further subclassified into KRAS amplifications, activating KRAS mutations, and KRAS switch II pocket mutations. KRAS amplifications typically involve the mutant G12C allele and are thought to drive resistance by increasing the abundance of the target protein beyond the inhibitory capacity of KRAS^{G12C} OFF state inhibitors, effectively overwhelming drug engagement.^{92,93} In contrast, secondary activating KRAS mutations—most commonly at codons G12, G13, or Q61 (e.g., KRAS^{G12D}, ^{G13D}, or ^{Q61H})—typically arise *in trans* (i.e., on the wild-type allele) and are not susceptible to mutant selective KRAS^{G12C} inhibitors, thereby restoring oncogenic KRAS signaling through alternative activating alleles. Notably, novel classes of KRAS inhibitors with broader target coverage, such as KRAS isoform selective (“pan-KRAS”) and RAS-multi (“pan-RAS”) inhibitors, discussed in more detail later in discussion, have been shown to effectively overcome second hit activating KRAS mutations in preclinical models.^{90,95,96} KRAS switch II pocket mutations, which invariably occur *in cis* with the G12C

mutation and most frequently involve residues R68, M72, H95, and Y96, drive resistance by inducing conformational changes of the KRAS protein that specifically prevent binding of inhibitors targeting the switch II pocket.^{92,97} This is particularly important as novel KRAS inhibitors that bind through alternative mechanisms have been shown to retain activity despite these mutations.^{90,97,98} Acquired off-target resistance alterations beyond KRAS most commonly affect other signaling nodes of the RAS/MAPK cascade. These include upstream regulators such as receptor tyrosine kinase (RTK) amplifications, fusions, or mutations; alterations in the paralogous RAS isoforms NRAS and HRAS; and, most frequently, activating events in downstream effectors such as BRAF and MAP2K1. Importantly, a recent pooled analysis of paired pre-treatment and post-progression ctDNA sequencing data from 143 patients treated with various KRAS^{G12C} OFF state inhibitors revealed that approximately one in four patients developed multiple concurrent RAS/MAPK pathway alterations during treatment, with the highest incidence observed in CRC and a lower frequency in NSCLC. Notably, these cases often involved the co-emergence of on-target KRAS mutations alongside off-target alterations in other RAS/MAPK pathway genes, underscoring the molecular complexity of genetic resistance and highlighting the need for therapeutic strategies capable of targeting multiple parallel resistance mechanisms.⁹⁰ While RAS/MAPK pathway alterations have been well characterized as bona fide resistance drivers in preclinical models, the functional relevance of other, less frequently observed acquired gene alterations—such as *PIK3CA* mutations, *MYC* amplifications, and *TP53* loss-of-function events—remains incompletely understood and warrants further rigorous mechanistic investigation.

Non genetic mechanisms of resistance

Although genomic alterations in the RAS/MAPK pathway are well-established contributors to acquired resistance to KRAS^{G12C} inhibitors, approximately 25% of patients experience disease progression without detectable genetic drivers, implicating non-genetic mechanisms as key contributors of therapeutic escape.⁸⁴ These heterogeneous mechanisms are less well defined, owing in part to the limited availability of longitudinal tumor biopsies—essential for capturing dynamic epigenetic and transcriptional adaptations in both tumor cells and the surrounding microenvironment—and the inherent challenges of modeling such complexity in preclinical systems. One of the most well-characterized forms of non-genetic resistance is adaptive resistance, referring to the rapid reactivation of RAS/MAPK signaling following KRAS^{G12C} inhibition, that is driven by relief of negative feedback loops that normally suppress upstream signaling. Preclinical studies have shown that the resulting loss of negative feedback control leads to the activation of various receptor tyrosine kinases (RTKs), including EGFR, HER2, FGFR, and c-MET. This RTK activation has two main effects. First, it can activate wild-type RAS isoforms (i.e., NRAS and HRAS), thereby circumventing the inhibited mutant KRAS^{G12C}. Second, RTK signaling into KRAS^{G12C} itself can shift the equilibrium further toward the GTP-bound on-state.^{74,76,99,100} In parallel, newly synthesized KRAS^{G12C} protein may evade inhibition by rapidly transitioning to the GTP-bound ON conformation, rendering OFF state inhibitors unable to effi-

ciently bind and block signaling.¹⁰¹ In CRC, where RTK signaling and adaptive resistance is particularly prominent, EGFR has been shown to be the dominant driver of downstream signaling reactivation.¹⁰⁰ These insights, along with prior clinical experience in BRAF^{V600E}-mutant CRC, where vertical pathway inhibition through combined BRAF and EGFR blockade significantly improved outcomes, provided the rationale for evaluating KRAS^{G12C} inhibitors in combination with anti-EGFR antibodies.⁷⁷ Notably, in patients with CRC this approach has led to a doubling of ORRs compared to single agent KRAS^{G12C} inhibition, as discussed in more detail above.³¹ In tumor types other than CRC, RTK-mediated adaptive resistance appears more heterogeneous, posing a significant challenge for the development of effective combination treatment strategies. Novel bi- and multi-specific antibodies targeting various RTKs, along with SOS1 and SHP2 inhibitors that act on convergent downstream nodes, represent promising combination partners for KRAS inhibitors to overcome adaptive resistance.

While canonical RAS/MAPK pathway reactivation remains the best-characterized and clinically validated mechanism of adaptive resistance, emerging data highlight additional, orthogonal pathways that bypass KRAS^{G12C} inhibition. Notably, YAP/TAZ/TEAD signaling, which represents a central component of the Hippo pathway, has been increasingly recognized as a mediator of adaptive resistance by driving transcriptional programs that promote KRAS-independent tumor growth.^{83,84} Adachi et al. demonstrated that KRAS^{G12C} inhibition disrupts MAPK signaling-dependent membrane localization of the apical polarity protein Scribble, thereby attenuating Scribble/SHOC2/PP1c complex formation. This shift permits the nuclear translocation of YAP, which activates TEAD mediated transcription of YAP target genes, including the RAS superfamily protein MRAS. Following RTK-mediated activation, membrane-localized MRAS forms a complex with SHOC2/PP1c that dephosphorylates CRAF, ultimately leading to MAPK pathway reactivation.¹⁰² Importantly, another study has shown that combining sotorasib with a novel pan-TEAD inhibitor, which blocks interactions between the YAP-TAZ protein complex and all four paralogs of TEAD proteins, exhibits highly synergistic antitumor activity *in vivo*, supporting the therapeutic potential of co-targeting YAP/TEAD signaling to overcome adaptive resistance to KRAS^{G12C} inhibition.¹⁰³ Notably, YAP/TAZ signaling is also implicated in promoting epithelial-mesenchymal transition (EMT), a process by which epithelial cells lose apical-basal polarity and cell-cell adhesion, acquiring mesenchymal properties such as enhanced motility and invasiveness. This cellular plasticity, often driven by transcription factors such as ZEB1, SNAI1, SNAI2, and TWIST, has been increasingly recognized as a key mechanism of resistance to targeted therapies, including KRAS^{G12C} and KRAS^{G12D} inhibitors.^{91,104} In addition to EMT, histologic transformation represents another form of lineage plasticity that has been linked to driving resistance to KRAS^{G12C} inhibitors in NSCLC. This includes transitions from adenocarcinoma to squamous cell carcinoma, as well as shifts toward alveolar or mucinous differentiation states, all of which are thought to reflect underlying transcriptional reprogramming that reduces tumor dependency on KRAS signaling.^{92,105,106}

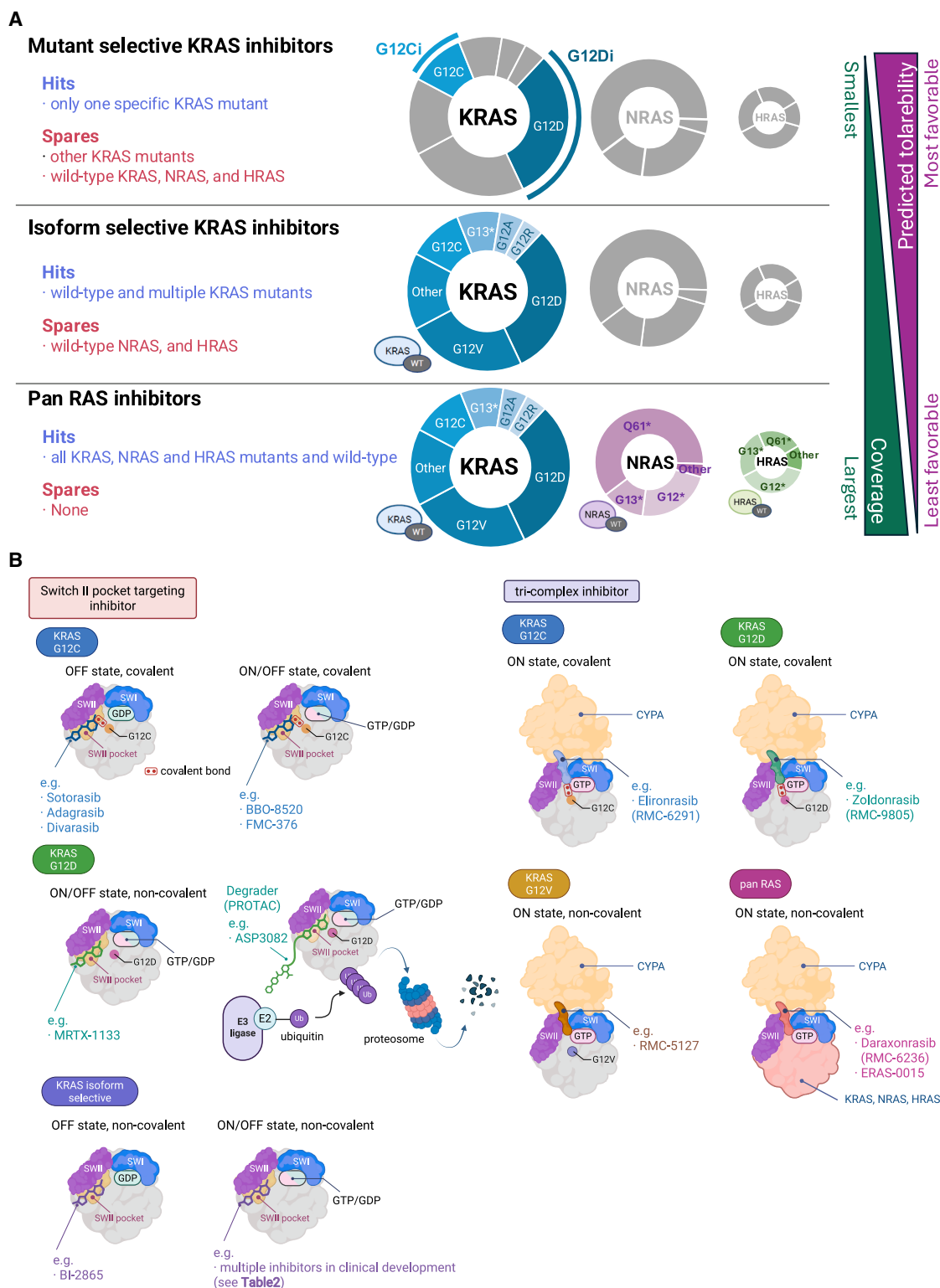


Figure 3. Emerging KRAS inhibitors: classification and mechanisms of action

(A) Classification of KRAS inhibitors and their spectrum of activity (mutant selective vs. pan-RAS). Schematic classification of RAS inhibitors based on target selectivity. Mutant-selective inhibitors are designed to target a single KRAS mutant (e.g., KRAS G12C or G12D), while sparing other KRAS mutants and wild-type KRAS, NRAS, and HRAS. Isoform-selective inhibitors broaden their activity to multiple KRAS mutants as well as wild-type KRAS, but spare NRAS and HRAS. In contrast, pan-RAS inhibitors act across all RAS isoforms (KRAS, NRAS, and HRAS), targeting both mutant and wild-type proteins. The

(legend continued on next page)

EMERGING LANDSCAPE OF NOVEL (K)RAS INHIBITORS

Although switch II pocket targeting KRAS^{G12C} OFF state inhibitors such as sotorasib and adagrasib have demonstrated clinical efficacy, their therapeutic durability is limited by the rapid emergence of resistance.¹⁰⁷ This has spurred the development of next-generation KRAS^{G12C} inhibitors aimed at achieving improved potency, greater target engagement, and more sustained ON target inhibition. Nonetheless, KRAS^{G12C} accounts for only about 3% of all KRAS mutations across cancers, rendering the vast majority of patients with KRAS-driven tumors ineligible for these agents (Figure 1B). This underscores the urgent need for KRAS inhibitors capable of targeting more prevalent mutations such as G12D and G12V. The biochemical properties of KRAS mutants vary considerably, presenting distinct challenges for drug development. KRAS^{G12C} retains intrinsic GTP hydrolysis activity, allowing it to cycle between GTP- and GDP-bound states.¹⁵ This cycling makes the GDP-bound OFF state relatively accessible, facilitating the design of covalent inhibitors targeting the mutant cysteine residue. In contrast, KRAS^{G12D} involves substitution with aspartate, a negatively charged and polar amino acid with a relatively small and flexible side chain. This reduced steric hindrance causes less distortion to the local protein structure compared to bulkier residues, allowing for small molecule engagement despite the absence of a nucleophilic cysteine residue required for covalent binding. However, the KRAS^{G12D} mutation significantly decreases intrinsic and GAP-mediated GTP hydrolysis, locking the protein predominantly in the active GTP-bound state.¹⁵ The altered electrostatic environment and persistent activation make KRAS^{G12D} a more challenging target for conventional OFF state inhibitors, necessitating alternative strategies such as ON state inhibitors.¹⁰⁸ KRAS^{G12V}, the second most common oncogenic mutation, replaces glycine with valine, a bulky and hydrophobic amino acid. The large side chain introduces considerable steric hindrance and conformational rigidity, especially in the switch regions, which further suppresses GTP hydrolysis and favors the active ON state.¹⁵ The lack of a reactive cysteine residue and other suitable functional groups poses a significant challenge to the design of inhibitors capable of effectively engaging KRAS^{G12V}. These distinct structural and biochemical properties of KRAS mutants necessitate tailored approaches in drug development. In response to these challenges, the landscape of KRAS targeted therapies is rapidly evolving. Based on target selectivity emerging classes of novel (K)RAS inhibitors can be broadly categorized into three groups: 1) mutant-selective inhibitors, which target a specific (K)RAS mutation (e.g., KRAS^{G12C}/KRAS^{G12D}) while sparing all wild-type RAS isoforms; 2) isoform-selective inhibitors that target multiple KRAS mutants (also called “pan-

KRAS inhibitors”) and wild-type KRAS while minimizing activity against NRAS and HRAS; and 3) and pan-RAS inhibitors, which exhibit the broadest coverage by targeting mutant and wild-type forms across all RAS isoforms (Figure 3A). While mutant-selective inhibitors offer the advantage of high specificity and potentially reduced toxicity, they are more prone to resistance due to pathway bypass or secondary mutations. In contrast, pan-RAS inhibitors may provide more durable target suppression and broader efficacy but carry a higher risk of on-target toxicity in normal tissues due to pan wild-type RAS inhibition. Isoform selective KRAS inhibitors could potentially offer a favorable balance, covering >90% of patients with KRAS mutant cancer, while limiting toxicity through reduced activity against non-KRAS isoforms. Functionally, novel (K)RAS inhibitors can be further subclassified based on their binding mode into switch II pocket targeting inhibitors and tri-complex-forming agents. In addition, available agents vary in their ability to inhibit RAS in its GTP-bound ON state (Figure 3B). Beyond traditional small-molecule inhibitors, targeted protein degradation approaches such as PROTACs are under investigation to eliminate mutant KRAS proteins from cells. In the next section, we provide a comprehensive overview of emerging KRAS-targeted therapeutic strategies beyond existing KRAS^{G12C} OFF state inhibitors, putting a particular emphasis on both preclinical and preliminary clinical data supporting these novel approaches.

MUTANT SELECTIVE INHIBITORS

KRAS^{G12C} selective ON state inhibitors

KRAS^{G12C} exhibits the highest intrinsic GTP hydrolysis rate among KRAS mutants; nevertheless, most of the protein remains in the active, GTP-bound ON state. This supports the rationale for directly targeting KRAS^{G12C} in its GTP-bound ON state conformation.

Elironrasib (RMC-6291) is a first-in-class covalent tri-complex ON state inhibitor that selectively targets the active GTP-bound form of RAS^{G12C}. The compound initially forms a binary complex with the intracellular chaperone cyclophilin A (CYPA), which subsequently engages RAS to assemble a stable tri-complex.⁹⁸ This structure-based strategy, inspired by natural products such as rapamycin and cyclosporin, leverages the high intracellular abundance of CYPA and its favorable electrostatic complementarity with the effector-binding surface of RAS. Notably, elironrasib is a synthetic derivative of Sanglifehrin A, a natural product known for its high-affinity binding to CYPA. Within the tri-complex, CYPA occupies the switch I and switch II motifs of RAS, thereby sterically hindering the binding of multiple downstream effectors. Selectivity for KRAS^{G12C} is conferred through the covalent modification of the reactive thiol group uniquely

spectrum of activity ranges from the narrowest (mutant-selective) to the broadest (pan-RAS), with implications for therapeutic efficacy, predicted tolerability, and potential on-target toxicity.

(B) Mechanisms of action of novel RAS inhibitors (switch II pocket targeting inhibitors vs. tri-complex inhibitors). Schematic overview of novel RAS inhibitors highlighting key differences in binding modes, mechanisms of action, conformational state selectivity, and target specificity. Switch II pocket inhibitors include covalent OFF-state inhibitors of KRAS^{G12C} (e.g., sotorasib, adagrasib, and divarasil), as well as emerging covalent agents capable of binding KRAS^{G12C} in both ON and OFF states (e.g., BBO-8520 and FMC-376). For KRAS^{G12D}, switch II pocket inhibitors include ON/OFF-state inhibitors (e.g., MRTX-1133) and degraders (e.g., ASP3082). KRAS isoform-selective switch II pocket inhibitors have also been developed, targeting either the OFF state alone (e.g., BI-2865) or both ON and OFF states (multiple agents in development). In contrast, tri-complex inhibitors act by targeting the ON state of KRAS through the formation of a tri-complex with cyclophilin A (CYPA). Examples include elironrasib (RMC-6291, covalent G12C-selective), zoldonrasib (RMC-9805, covalent G12D-selective), RMC-5127 (non-covalent G12V-selective), as well as daraxonrasib (RMC-6236) and ERAS-0015 as pan-RAS inhibitors.

introduced by the oncogenic cysteine substitution at codon 12. This distinctive tri-complex inhibitor (TCI) mechanism enables potent and selective suppression of oncogenic RAS signaling.¹⁰⁹ Importantly, RM-018, a preclinical analog of RMC-6291, has been shown to overcome KRAS switch II pocket resistance mutations, such as KRAS^{G12D}, which are known to confer resistance to switch II pocket targeting KRAS^{G12C} OFF state inhibitors.^{97,98} In a phase 1 clinical trial, elironasib has demonstrated encouraging preliminary clinical tolerability and efficacy in patients with KRAS^{G12C} mutant NSCLC and CRC, including an ORR of 50% in patients with NSCLC previously treated with KRAS^{G12C} OFF state inhibitors (Table 2).¹¹⁰ Based on these data, elironasib was recently granted breakthrough therapy designation by the FDA for the treatment of patients with advanced KRAS^{G12C}-mutant NSCLC who have received standard first-line therapies.

BBO-8520 is a first-in-class covalent dual inhibitor that covalently binds to the switch II pocket of KRAS^{G12C} and directly targets both its GTP-bound ON and GDP-bound OFF states. Upon binding, the switch I region adopts an open conformation in both structural states, suggesting that BBO-8520 forces KRAS^{G12C} in the ON state into a signaling-incompetent conformation.¹²⁰ Importantly, preclinical studies have demonstrated that BBO-8520 retains potent antitumor activity in models resistant to KRAS^{G12C} OFF state inhibitors, including tumors with KRAS^{G12C} amplification and receptor tyrosine kinase overexpression, highlighting its potential to overcome acquired resistance. BBO-8520 is currently being evaluated in a phase 1 clinical trial in patients with KRAS G12C-mutant NSCLC. FMC-376 is another selective, orally bioavailable covalent inhibitor that targets both ON and OFF states of KRAS^{G12C} and has been shown to overcome resistance to first-generation inhibitors in preclinical models. It has demonstrated efficacy in patient derived xenograft models of NSCLC, PDAC, and CRC, and is currently being evaluated in a Phase 1/2 clinical trial.¹²¹

KRAS^{G12D}-selective inhibitors

KRAS^{G12D} mutations account for approximately 28% of all KRAS mutations and are most frequently observed in CRC and PDAC (Figure 1B). The absence of a reactive cysteine residue in KRAS^{G12D} presents a significant challenge for the development of covalent inhibitors. Consequently, initial therapeutic efforts were primarily focused on the design and optimization of non-covalent inhibitors targeting this mutation.

MRTX1133 is the first potent and selective noncovalent inhibitor targeting KRAS^{G12D}. It was rationally designed based on structural insights into the conserved binding interactions within the switch II pocket across KRAS variants, and on the principle that a positively charged amine moiety positioned adjacent to the acidic side chain of Asp12 enables selective molecular interaction. Notably, MRTX1133 binds both the inactive GDP-bound and active GTP-bound states of KRAS^{G12D}. Trapping the mutant protein in its GDP-bound state locks it in an inactive conformation, whereas binding to the GTP-bound state induces conformational changes in the switch I and II regions that impairs KRAS function. Since these loops are critical for interactions with downstream effector proteins, the induced conformational changes likely impair the ability of KRAS^{G12D} to engage effectors in both nucleotide states.¹²² MRTX1133 demonstrated strong antitumor activity in KRAS^{G12D}-mutant xenograft models. How-

ever, its phase 1/2 clinical trial was terminated after the completion of phase 1 due to highly variable and suboptimal pharmacokinetics.

Zoldonrasib (RMC-9805) is a first-in-class covalent tri-complex inhibitor that selectively targets the ON state of KRAS^{G12D} by forming a stable complex with KRAS and cyclophilin A. Zoldonrasib achieves selective covalent bonding with the Asp12 residue of KRAS^{G12D} through precise positioning of a reactive aziridine warhead within the switch II pocket, which alters the local microenvironment to facilitate bond formation and effectively inhibits KRAS signaling. Zoldonrasib demonstrated potent *in vivo* antitumor activity and a favorable pharmacokinetic profile, supporting once-daily oral dosing in preclinical models of KRAS^{G12D}-mutant PDAC, NSCLC, and CRC.¹²³ Notably, preliminary data from a phase I trial of zoldonrasib in patients with KRAS^{G12D}-mutant solid tumors demonstrated encouraging antitumor activity and a favorable safety profile in pretreated, advanced NSCLC and PDAC, with a reported ORR of 61% and 30%, respectively.^{111,112} Promising preliminary clinical activity and tolerability data have also been recently presented for GFH375, a mutant-selective KRAS^{G12D} inhibitor that targets both the GTP-bound ON and inactive GDP-bound OFF states of the protein.¹²⁴ In detail, the ORR in patients with advanced, pretreated NSCLC and PDAC were 58% and 41%, respectively.^{113,114} Recently, early phase I clinical data were also reported for HRS-4642 and INCB161734, two others dual-state KRAS^{G12D} inhibitors.^{125,126} At the recommended phase II dose levels HRS-4642 demonstrated ORRs of 33% in NSCLC and 25% in PDAC,¹¹⁵ while INCB161734 showed an ORR of 34% in PDAC¹¹⁶ (Table 2).

ASP3082 is the first PROTAC molecule designed to target KRAS^{G12D}, harnessing the cell's ubiquitin-proteasome system to selectively degrade this oncogenic protein. As a proteolysis-targeting chimera (PROTAC), ASP3082 induces targeted protein degradation rather than conventional inhibition, offering a novel mechanism of action.¹²⁷ Recently presented preliminary phase I trial data of ASP3082 in KRAS^{G12D}-mutant solid tumors demonstrated a favorable safety profile and encouraging antitumor activity with an ORR of 38% in patients with pretreated NSCLC.¹¹⁷ Preliminary functional data have been reported for several other KRAS^{G12D}-targeting agents, including LY3962673, QTX3046, AZD0022, GFH375, INCB161734, and TSN1611, all of which are currently being evaluated in early-phase clinical trials.

Mutant selective inhibitors beyond KRAS^{G12C/D}

Currently, drug development efforts targeting specific KRAS mutations beyond G12C and G12D remain limited. Given that KRAS^{G12V} represents the second most prevalent KRAS alteration overall (Figure 1B), there is an urgent unmet need to advance therapeutic strategies specifically directed against KRAS^{G12V}-driven malignancies. Recently, the discovery of the first mutant selective KRAS^{G12V} inhibitor was reported. RMC-5127 is a potent, orally bioavailable, and selective noncovalent inhibitor that targets the active, GTP-bound form of KRAS^{G12V} through a tri-complex mechanism. By forming a stable tri-complex with KRAS^{G12V} in its GTP bound ON conformation and cyclophilin A (CypA), RMC-5127 induces the rapid disruption of downstream effector interactions via steric occlusion, effectively suppressing downstream signaling.¹²⁸ Additional mutant

Table 2. Developmental status and preliminary clinical efficacy of novel classes of (K)RAS inhibitors

Drug/Company	Target	Mechanism of action	Phase of development	Trial/status	Preliminary clinical efficacy data	Reference
Mutant selective KRAS inhibitors						
Elironrasib (RMC-6291)/ Revolution Medicines	KRAS ^{G12C}	ON state inhibitor Molecular glue (Tricomplex) Covalent	Phase 1/2	NCT05462717 (active, not recruiting) NCT06128551 (recruiting) NCT06162221 (recruiting)	NSCLC KRAS ^{G12C} i naive (<i>n</i> = 7): ORR 43%, DCR 100% KRAS ^{G12C} i pretreated (<i>n</i> = 10): ORR 50%, DCR 100% CRC KRAS ^{G12C} i naive (<i>n</i> = 20): ORR 40%, DCR 80%	Jänne et al. ¹¹⁰
FMC-376/Frontier Medicines	KRAS ^{G12C}	ON/OFF state inhibitor Covalent	Phase 1	NCT06244771 (recruiting)	/	/
BBO-8520/BridgeBio Oncology Therapeutics	KRAS ^{G12C}	ON/OFF state inhibitor Switch-2 pocket Covalent	Phase 1	BBO-8520	/	/
Zoldonrasib (RMC-9805)/ Revolution Medicines	KRAS ^{G12D}	ON state inhibitor Molecular glue (Tricomplex) Covalent	Phase 1/2	NCT06040541 (recruiting) NCT06445062 (recruiting) NCT06162221 (recruiting)	NSCLC (<i>n</i> = 18) ORR 61%, DCR 89% PDAC (<i>n</i> = 40) ORR 30%, DCR 80%	Arbor et al. ¹¹¹ ; Spira et al. ¹¹²
GFH375 (VS-7375)/V erastem Oncology	KRAS ^{G12D}	ON/OFF state inhibitor Noncovalent	Phase 1/2	NCT06500676 (recruiting)	NSCLC (<i>n</i> = 26) ORR: 58%, DCR: 89% PDAC (<i>n</i> = 59) ORR: 41%, DCR: 97% mPFS: 5.5 months (95% CI: 4.3–7.2)	Zhou et al. ¹¹³ ; Lu et al. ¹¹⁴
HRS-4642/Jiangsu Hengrui Pharmaceuticals	KRAS ^{G12D}	ON/OFF state inhibitor Noncovalent	Phase 1/2	NCT06520488 (recruiting)	NSCLC (<i>n</i> = 9) ORR: 33%, DCR: 89% mPFS 8.4 months (95% CI: 1.4–nr) PDAC (<i>n</i> = 12) ORR: 25%, DCR: 100% mPFS 4.4 months (95% CI: 2.6–7.2)	Xiong et al. ¹¹⁵
INCB161734/Incyte Corporation	KRAS ^{G12D}	ON/OFF state inhibitor Switch-2 pocket Noncovalent	Phase 1	NCT06179160 (recruiting)	PDAC (<i>n</i> = 29) ORR: 34%, DCR: 86%	Desai et al. ¹¹⁶
ASP3082/Astellas Pharma Inc	KRAS ^{G12D}	PROTAC	Phase 1	NCT05382559 (recruiting)	NSCLC (<i>n</i> = 24) ORR: 38%, DCR: 92%	Spira et al. ¹¹⁷
MRTX1133/Mirati Therapeutics Inc	KRAS ^{G12D}	OFF state inhibitor Switch-2 pocket Noncovalent	Phase 1	NCT05737706 (terminated)	/	/

(Continued on next page)

Table 2. Continued

Drug/Company	Target	Mechanism of action	Phase of development	Trial/status	Preliminary clinical efficacy data	Reference
HRS-4642/Jiangsu HengRui Medicine Co.	KRAS ^{G12D}	ON/OFF state inhibitor Noncovalent	Phase 1	NCT05533463 (recruiting)	/	/
LY3962673/Lilly	KRAS ^{G12D}	OFF state inhibitor Noncovalent	Phase 1	NCT06586515 (recruiting)	/	/
QTX3046/Quanta Therapeutics	KRAS ^{G12D}	ON/OFF state inhibitor Noncovalent	Phase 1	NCT06428500 (active, not recruiting)	/	/
AZD0022/AstraZeneca	KRAS ^{G12D}	ON/OFF state inhibitor Switch-2 pocket Noncovalent	Phase 1	NCT06599502 (recruiting)	/	/
INCB161734/Incyte Corporation	KRAS ^{G12D}	ON/OFF state inhibitor Switch-2 pocket Noncovalent	Phase 1	NCT06179160 (recruiting)	/	/
TSN1611/Tyligand Pharmaceuticals	KRAS ^{G12D}	ON/OFF state inhibitor Noncovalent	Phase 1	NCT06385925 (recruiting)	/	/
RMC-5127 Revolution Medicines	KRAS ^{G12V}	ON state inhibitor Molecular glue (Tricomplex) Noncovalent	Preclinical	/	/	/
Isoform selective KRAS inhibitors (“pan KRAS inhibitors”)						
BI-3706674/Boehringer Ingelheim	Multiple KRAS mutations/KRAS wildtype amplification	OFF state inhibitor Switch-2 pocket Noncovalent	Phase 1	NCT06056024 (Active, not recruiting)	/	/
LY4066434/Lilly	Multiple KRAS mutations	OFF state inhibitor Noncovalent	Phase 1	NCT06607185 (recruiting)	/	/
ALTA-3263/Alterome	Multiple KRAS mutations	ON/OFF state Noncovalent	Phase 1	NCT06835569 (recruiting)	/	/
PF-07985045/Pfizer	Undisclosed	Undisclosed	Phase 1	NCT06447662 (recruiting)	/	/
PF-07934040/Pfizer	Undisclosed	Undisclosed	Phase 1	NCT06704724 (recruiting)	/	/
QTX3034/Quanta	Multiple KRAS mutations (KRAS G12D preferring)	OFF state inhibitor Noncovalent	Phase 1	NCT06227377 (recruiting)	/	/
QTX3544/Quanta	Multiple KRAS mutations (KRAS G12V preferring)	OFF state inhibitor Noncovalent	Phase 1	NCT06715124 (recruiting)	/	/
AMG410/Amgen	Multiple KRAS mutations	ON/OFF state inhibitor Switch-2 pocket Noncovalent	IND application filed	NCT07094113 (not yet recruiting)	/	/

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Table 2. Continued

Drug/Company	Target	Mechanism of action	Phase of development	Trial/status	Preliminary clinical efficacy data	Reference
ERAS-4001/Erasca	Multiple KRAS mutations	ON/OFF state inhibitor Molecular glue (Tricomplex)	Phase 1	NCT07021898 (recruiting)	/	/
BBO-11818	Multiple KRAS mutations	ON/OFF state inhibitor Noncovalent	Phase 1	NCT06917079 (recruiting)	/	/
Multi RAS inhibitors ("pan RAS inhibitors")						
Daraxonrasib (RMC-6236) Revolution Medicines	Multiple KRAS, NRAS and HRAS mutations	ON state inhibitor Molecular glue (Tricomplex) Noncovalent	Phase 3	NCT06625320 (recruiting) NCT06881784 (recruiting) NCT05379985 (recruiting) NCT06128551 (recruiting) NCT06040541 (recruiting) NCT06445062 (recruiting) NCT06162221 (recruiting)	NSCLC with KRAS^{G12X} mutations (n = 42) ORR 38%, DCR 100% mPFS: 9.8 months (95% CI: 6–12.3) mOS: 17.7 months (95% CI: 13.7–ne) PDAC with KRAS^{G12X} mutations (n = 42) ORR 29%, DCR 91% mPFS: 8.5 months (95% CI: 5.3–11.7) PDAC with other RAS mutations (n = 57) ORR 25%, DCR 93% mPFS: 7.6 months (95% CI: 5.9–11.1)	Garrido-Laguna et al. ¹¹⁸ ; Punekar et al. ¹¹⁹
ERAS-0015/Erasca	Multiple KRAS, NRAS and HRAS mutations	ON state inhibitor Molecular glue (Tricomplex)	Phase 1	NCT06983743 (recruiting)	/	/

Abbreviations: DCR – disease control rate; ORR – objective response rate; mPFS – median progression free survival; mOS – median overall survival; Ref. – reference; NSCLC – non-small cell lung cancer; CRC – colorectal cancer; PDAC – pancreatic ductal adenocarcinoma; KRAS^{G12C}i – KRAS^{G12C} inhibitor.

selective ON-state tri-complex inhibitors, including RMC-0708 (RM-046) targeting KRAS^{Q61H} and RMC-8839 targeting KRAS^{G13C}, as well as covalent inhibitors targeting the switch II pocket of KRAS^{G12S} are currently in preclinical development.^{129–131}

Multi RAS inhibitors (“pan RAS” inhibitors)

Therapeutic agents that target both mutant and wild-type forms of KRAS, NRAS, and HRAS offer the broadest potential clinical coverage, with an estimated 3.4 million new cases of RAS-mutant cancers globally each year.¹ Moreover, such pan-RAS inhibitors may provide an advantage in preventing adaptive resistance mediated by wild-type RAS isoforms (Figure 3A). However, there are theoretical concerns regarding on-target toxicity, as *in vivo* studies have shown that complete RAS knockout results in embryonic lethality in mice.

RMC-7977 (a tool compound) and daraxonrasib (RMC-6236) represent first-in-class, multi-RAS ON state inhibitors that function through a CYPA-dependent molecular glue mechanism. These compounds form non-covalent tri-complexes with active, GTP-bound RAS and cyclophilin A, thereby disrupting RAS-effector interactions. Notably, CYPA and these compounds occupy the switch I and II regions of RAS, sterically and directly obstructing the binding of downstream effectors such as RAF, PI3K, and RALGDS. This spatial interference leads to effective blockade of effector binding and potent inhibition of RAS signaling.¹³²

Distinct from other tri-complex inhibitors, this mechanism broadly targets multiple RAS isoforms—including KRAS, NRAS, and HRAS—and selectively acts on the active GTP-bound ON state. Importantly, daraxonrasib has been further shown to stimulate GTP hydrolysis, converting KRAS from its active GTP-bound form to the inactive GDP-bound form, a property particularly pronounced in KRAS^{G12X} mutants.¹³³ Extensive preclinical studies have demonstrated broad-spectrum anti-tumor activity across various RAS-mutant *in vivo* models, with especially strong activity in tumors harboring KRAS^{G12X} mutations. Moreover, daraxonrasib has been shown to overcome different mechanisms of acquired resistance to KRAS^{G12C} OFF state inhibitors, including secondary KRAS mutations, NRAS mutations, and upstream RTK amplification, mutation, or fusion events.⁹⁶ Notably, despite its broad target coverage including wild-type RAS, daraxonrasib was well tolerated in mice at therapeutically effective dose levels, potentially related to its preferential engagement of mutant GTP-bound RAS and reduced activity in normal tissue where RAS exists primarily in its inactive GDP-bound OFF state.⁹⁶ Preliminary results from the phase I clinical trial evaluating daraxonrasib in patients with advanced RAS mutant cancers demonstrated promising efficacy and a manageable safety profile in patients with previously treated, advanced NSCLC and PDAC harboring diverse (K)RAS mutations. In the subset of patients with KRAS^{G12X} mutant PDAC treated at the recommended dose of 300 mg per day, an ORR of 29% and a median PFS of 8.5 months were reported, leading to FDA breakthrough therapy designation for this indication. In patients with NSCLC, initial response rates of 38% and a median PFS of 9.8 months have been observed. Randomized phase III trials are currently underway comparing daraxonrasib to standard-of-care therapies in advanced PDAC and NSCLC

(Table 2). Notably, preliminary data on acquired resistance to daraxonrasib have identified acquired KRAS amplifications in 35% (9/26) of patients at clinical progression.¹³⁴ This finding is intriguing given that daraxonrasib has demonstrated retained activity in KRAS amplified preclinical models, warranting further clinical and functional validation. In parallel, Sang et al. demonstrated that KRAS^{Y64X} mutations confer on-target resistance to daraxonrasib through the attenuation of tri-complex formation, along with kinase-dead BRAF mutations that impair target engagement.¹³⁵

ERAS-0015 is another pan-RAS molecular glue currently in early clinical development. It selectively targets the GTP-bound ON state of both mutant and wild-type RAS and has demonstrated favorable antiproliferative activity across a range of RAS-mutant cell lines and potentially higher binding affinity for cyclophilin A.¹³⁶

Notably, unlike mutant-selective inhibitors, daraxonrasib has shown evidence of on-target RAS-MAPK toxicity in patients, including a classical acneiform skin rash previously seen with other MAPK pathway inhibitors, as well as gastrointestinal adverse events, including nausea, vomiting, and diarrhea. Although daraxonrasib has shown better tolerability than many other classes of inhibitors that produce non-selective RAS/MAPK pathway blockade (e.g., MEK, ERK, and SHP2 inhibitors), these observations do raise the possibility that on-target toxicity associated with pan-RAS inhibitors may ultimately limit the exposures and degree of RAS inhibition that can be achieved in patients. So far, the initial ORRs observed with daraxonrasib in NSCLC, PDAC, and CRC are lower than those observed with the most advanced mutant-selective inhibitors.^{118,119} Therefore, it is possible that strategies to achieve more selective inhibition of aberrant KRAS signaling may be needed to realize optimal clinical efficacy.

Isoform selective KRAS inhibitors (“pan KRAS inhibitors”)

A class of inhibitors selectively targeting multiple KRAS mutants along with wild-type KRAS—but not NRAS or HRAS—would cover approximately 90% of clinically observed RAS mutations. Given that mouse genetic studies, discussed above, suggest that individual RAS isoforms, including KRAS, are not required by normal cells due to compensatory signaling by other RAS isoforms, KRAS isoform-selective targeting may offer a more favorable toxicity profile compared to pan-RAS inhibitors. However, it remains to be demonstrated clinically whether KRAS-specific targeting will be tolerable in patients. Moreover, selective targeting of KRAS also raises the potential risk of resistance emerging through compensation by signaling through wild-type NRAS or HRAS isoforms. BI-2865 is a non-covalent pan-KRAS inhibitor that binds selectively and with high affinity to the inactive, GDP-bound OFF state of KRAS, while sparing NRAS and HRAS isoforms. It targets the switch II pocket, partially overlapping with the binding site of covalent KRAS^{G12C} inhibitors such as sotorasib and adagrasib. However, it does not form a covalent bond and does not extend into the P loop channel near the G12 residue.⁹⁵ Importantly, BI-2865 demonstrated broad *in vitro* and *in vivo* activity against various KRAS mutant models, including those harboring G12C, G12D, G12V, and A146V mutations, as well as KRAS-amplified tumors. In particular, cancer cell lines

with *KRAS* wild-type amplification (copy number >7) showed the greatest sensitivity to BI-2865 among all *KRAS*-altered models.¹³⁷ Since wild-type *KRAS* predominantly adopts an inactive, GDP-bound OFF state,¹³⁸ inhibitors that target the inactive OFF state of *KRAS* may be more effective than those targeting the active ON state. Consistent with these preclinical data BI-3706674, a clinical analogue derived from BI-2865, is currently being evaluated in a phase 1 clinical trial in patients with gastroesophageal cancers harboring either *KRAS* wild-type amplifications or *KRAS* G12V mutations (NCT06056024). Importantly, BI-2865 and BI-3706674 have been shown to lack activity against the GTP-bound ON state of *KRAS*. This pharmacologic limitation may reduce their effectiveness in tumors where the intrinsic GTP hydrolysis of *KRAS* mutants is impaired or where the equilibrium is shifted strongly toward the on-state. Reflecting this, limited activity of BI-2865 was observed in models harboring *KRAS*^{G12R} and *KRAS*^{Q61X} mutations, which possess extremely low intrinsic hydrolysis rates and are therefore insensitive to pure OFF-state inhibitors that target the GDP-bound form.⁹⁵ To address this challenge, next-generation *KRAS* isoform-selective inhibitors are being developed that can engage both the ON and OFF states, thereby expanding their potential therapeutic scope across a wider spectrum of *KRAS*-driven cancers (Table 2).

KEY CHALLENGES AND FUTURE DIRECTIONS IN RAS TARGETING

KRAS has long been considered a holy grail of cancer drug development due to the high prevalence of *KRAS* mutations across multiple tumor types and its historical reputation as an undruggable target.⁷ The breakthrough development of *KRAS*^{G12C} OFF state inhibitors has reinvigorated the field, marking a paradigm shift in RAS targeting and opening the door to broader RAS-directed strategies. As a result, (K)RAS-targeted therapies now hold the potential to transform the treatment landscape for a significant proportion of patients with cancer, potentially impacting more than 20% of all solid tumors.

However, despite this progress and the growing momentum, several critical challenges remain that must be addressed to fully realize the therapeutic potential of RAS inhibition. So far, clinical experience with approved first-generation *KRAS*^{G12C} OFF state inhibitors has demonstrated modest response rates and limited durability in many cases, falling short of the outcomes achieved with other targeted therapies. A critical next step in advancing *KRAS*^{G12C} inhibition is the development of best-in-class agents that combine high potency, favorable pharmacokinetics, and acceptable safety profiles. Early-phase data suggest that newer *KRAS*^{G12C} OFF state inhibitors—such as divarasil and D3S-001—with enhanced potency, improved selectivity, and stronger target engagement may induce deeper and more durable responses.^{35,43,53} Ongoing head-to-head phase 3 trials, such as KRASCENDO-1 (NCT06497556),⁶³ which compares divarasil to sotorasib or adagrasib in patients with *KRAS*^{G12C}-mutant NSCLC, will be critical to determine whether next-generation *KRAS*^{G12C} OFF state inhibitors offer superior efficacy over first-generation agents. However, despite improved biochemical properties, *KRAS*^{G12C} OFF state inhibitors as a class still face the fundamental question of whether selective inhibition of the

inactive GDP-bound state is sufficient to achieve sustained tumor control, particularly in the context of acquired resistance mechanisms that drive *KRAS* toward its active, GTP-bound conformation. Novel *KRAS*^{G12C} ON state inhibitors such as elironrasib or BBO-8520 may represent promising alternatives, though more mature clinical data are needed to fully evaluate their efficacy and tolerability.

Beyond *KRAS*^{G12C}, effective ON state coverage is anticipated to be even more imperative for achieving durable clinical responses, as other (K)RAS mutants exhibit markedly reduced intrinsic GTPase activity and are therefore predominantly locked in the GTP bound ON state.¹⁵ Importantly, novel mutant-specific *KRAS*^{G12D} ON state inhibitors such as zoldorasib and GFH375 have demonstrated encouraging antitumor activity in patients with PDAC and NSCLC with ORRs in the 30–60% range.^{111,112,139} However, it remains to be seen whether these responses are durable and translate into meaningful survival benefits, or if combination treatment strategies will be necessary to prevent or delay resistance. This may be particularly relevant in CRC, where RTK mediated adaptive pathway reactivation through wild-type RAS is expected to circumvent the inhibitory effects of mutant selective agents. Building on previous experiences with first generation *KRAS*^{G12C} OFF state inhibitors, anti-EGFR targeting antibodies may represent attractive combination partners in this setting.

Another appealing treatment strategy currently being explored in early phase clinical trials is to combine mutant-selective or isoform-selective *KRAS* inhibitors with pan-RAS inhibitors. This is based on the concept of exploiting the high therapeutic index of mutant-selective agents to potentially suppress the oncogenic driver while simultaneously blocking compensatory signaling through wild-type RAS isoforms or secondary (K)RAS mutations that may arise under selective pressure. Preliminary reports from ongoing phase I clinical trials combining the pan-RAS inhibitor daraxonrasib with either elironrasib (*KRAS*^{G12C} specific; NCT06128551) or zoldonrasib (*KRAS*^{G12D} specific; NCT06040541) support the feasibility of this concept, yet more mature data are needed to determine whether resistance can be effectively suppressed at clinically tolerable dose levels.¹⁴⁰ Combinations of tricomplex pan-RAS inhibitors with mutant-selective or isoform selective switch II pocket *KRAS* inhibitors may offer additional mechanistic advantages by engaging two non-overlapping sites on *KRAS* and coupling potent ON state inhibition from pan-RAS agents with the robust OFF state inhibition characteristic of most switch II pocket inhibitors. The potential of this combination strategy will need to be explored in future clinical trials.

Emerging classes of novel (K)RAS inhibitors with broader target coverage, including pan-RAS and isoform-selective *KRAS* inhibitors, have the potential to benefit a broad spectrum of patients across multiple cancer types (Figure 1B), including those with low-frequency variants, not amenable to mutant-selective inhibitors. Preliminary efficacy data in advanced, pre-treated PDAC and NSCLC are encouraging, with ORRs ranging from 25 to 38% and median PFS estimates between 7.6 and 9.8 months for the pan-RAS inhibitor daraxonrasib as monotherapy.^{118,119} While these results compare favorably to standard-of-care treatments in (K)RAS-mutant disease within this setting,^{141,142} there remains considerable room for improvement. In addition to the previously discussed concerns about

potential on-target toxicity associated with pan-RAS inhibitors, elucidating the mechanisms of resistance will be essential for advancing these therapies and designing rational combination strategies. Preliminary reports from studies evaluating post-progression ctDNA samples in patients treated with daraxonrasib suggest that newly acquired RAS/MAPK pathway alterations, including KRAS amplifications and KRAS^{Y64} mutations within the tricomplex inhibitor binding site (see Figure 1A) are frequent drivers of acquired genetic resistance.^{134,135} While such resistance mechanisms may be addressable through combinations of distinct classes of KRAS inhibitors leveraging different modes of action and binding properties, as discussed before, other recurrent alterations in genes downstream of RAS, such as BRAF and MAP2K1, will likely require alternative strategies such as novel concepts of vertical pathway inhibition. Beyond genetic mechanisms, deciphering the complex and largely uncharacterized landscape of non-genetic resistance mechanisms will be critical for the successful development of not only pan-RAS inhibitors but all classes of (K)RAS-targeted therapies. To address this challenge, on-treatment and post-progression tissue biopsies enabling comprehensive multi-omics characterization of both cancer cells and the tumor microenvironment under treatment pressure are indispensable. Based on experience with other targeted therapies, overcoming these non-genetic resistance mechanisms will likely require combination approaches, pairing (K)RAS inhibitors with chemotherapy, immunotherapy, or biological agents targeting critical downstream effectors or orthogonal pathways. Multiple of these concepts are currently explored preclinically, and some have already advanced to early phase clinical development.⁷¹

Ultimately, to increase response rates and, even more importantly, the durability of response, (K)RAS inhibitors will likely need to be deployed at earlier lines of therapy and disease stages, where clonal evolution and intratumoral heterogeneity that limit the activity of targeted agents are less advanced.¹⁴³ Eventually, novel RAS directed treatment strategies, such as cancer vaccines, may even be applied to prevent cancer formation from precursor lesions, yet achieving this ambitious vision will require sustained advances in our understanding of RAS biology, continued innovation in drug development, and carefully designed clinical studies.¹⁴⁴

CONCLUSION

The clinical development and approval of the mutant-selective KRAS^{G12C} inhibitors sotorasib and adagrasib have established that direct pharmacologic targeting of RAS, long considered undruggable, is clinically feasible. Yet, the modest durability of these agents underscores the need for more effective approaches to overcome resistance. Emerging classes of (K)RAS inhibitors with greater potency, broader target coverage, and distinct mechanisms of action hold the promise of extending benefit across a wider spectrum of RAS-driven cancers. Realizing this vision will require deeper insights into resistance mechanisms, predictive biomarkers to guide patient selection, and rational drug combinations that exploit complementary vulnerabilities. With such advances, the long elusive goal of effective and durable RAS inhibition could translate into lasting benefits for millions of patients with RAS-driven cancers worldwide.

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Doga C. Gulhan reports a patent application: SigMA software is freely available to academic researchers, and an international patent is pending.

All remaining authors have declared no conflicts of interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT 4.0 in order to assist with grammar and language clarity during the drafting process. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

SUPPLEMENTAL INFORMATION

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